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Video management

Abstract

The disclosure relates to a method of processing a sequence of image frames to reduce its length. One implementation may involve extracting coefficients (e.g., Discrete Cosine Transform coefficients) from components of individual frames, and comparing the resulting coefficients for sequential frames to identify frames having the least change from a prior frame. Also, scene change values for each frame may be calculated and placed in a sorted list to facilitate identification of frames for removal. Frame removal may be conducted in rounds, where a group of pictures (GOP) may only have one frame removed for any given round.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/158,531, filed Jan. 26, 2021, which is a continuation of the U.S. patent application Ser. No. 16/430,940, filed Jun. 4, 2019, now U.S. Pat. No. 10,951,959, which is a continuation of U.S. patent application Ser. No. 15/632,964 filed on Jun. 26, 2017, now U.S. Pat. No. 10,356,492, which is a continuation application of U.S. patent application Ser. No. 14/263,459 filed on Apr. 28, 2014, now U.S. Pat. No. 9,723,377, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Advances in data transmission technologies have allowed content providers to transmit multiple streams of content to users, and hundreds of channels of television programming can be delivered. In some cases, a user's viewing experience may automatically hop from one channel to another. For example, when a television program enters a commercial break, the ensuing commercials might actually be carried on a different channel or datastream, and the viewer's device (unknown to the viewer) may quickly switch tuning to the different channel for the duration of the commercial, and back to the television program (or to another channel carrying another commercial) when the commercial ends. To help tuners quickly lock on to the audiovisual signals during such rapid tuning, video transmission standards call for advertisements to begin with a few moments of blank/black video and silent audio. Unfortunately, many advertisers provide commercials that lack these moments of blank/black video and silent audio. Adding such moments to the beginning and end of the commercial may result in extending the commercial's duration, which may make it difficult for the commercial to fit within its allotted time in a commercial break. There remains a need to gracefully make these commercials comply with the video transmission standards while also allowing them to fit within their allotted time in a commercial break.

SUMMARY

(2) The following summary is for illustrative purposes only, and is not intended to limit or constrain the detailed description.

(3) Features herein relate to managing a video content comprising a sequence of image frames by dropping frames in the content. The number of frames to be dropped can depend on the amount of time that is desired to be trimmed from the content. The selection of frames to be dropped can begin with generating a frame value (e.g., a zero frequency value) for each frame in the video. The frame value can be the DC coefficient component (e.g., a zero frequency, or top-left element in a transform coefficient array) from, for example, each 8×8 pixel block extracted from the frame after a process such as a Discrete Cosine Transform. In some embodiments, the coefficients selected may be from the luminance (e.g., luma) component of the image frame, although chrominance components may be used if desired.

(4) When the frame values are generated for the frames in the video content, scene change values may then be generated for each frame in the video by comparing frame values of neighboring sequential frames. The scene change value for a frame may represent, for example, how much changed (e.g., visually) between the frame and its previous frame in the video content. For example, the scene change values may be determined as follows:

(5)
$$C(v) = \frac{1}{h} \sum_{n=1}^h \frac{1}{w} \sum_{m=1}^w \text{abs}[DC_v(m, n) - DC_u(m, n)]$$
 wherein $C(v)$ =the scene change value for the v th frame in the video content, $DC_v(m, n)$ =the Discrete Cosine Transform DC component of the m, n 8×8 block of DCT coefficients in the v th frame of the video content, $DC_u(m, n)$ =the Discrete Cosine Transform DC component of the m, n 8×8 block of DCT coefficients in the u th frame of the video content, $v=u+1$, w =the width of the frame, which may be measured in 8×8 pixel blocks, and h =the height of the frame, which may be measured in 8×8 pixel blocks.

(6) In some embodiments, the frame removal may be conducted in rounds, where each group of pictures (GOP) in the video may be limited to having just one (or other predetermined limit) frame

removed per round. The selection of frames for removal may generally seek to remove frames that have the least amount of change from a prior frame.

(7) In some embodiments, the video may initially include dependent frames, and dependent macroblock portions of frames. The present system may initially process these frames to recover their frame values in independent form, prior to calculating the scene change values and selecting frames for removal.

(8) The summary here is not an exhaustive listing of the novel features described herein, and are not limiting of the claims. These and other features are described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, claims, and drawings. The present disclosure is illustrated by way of example, and not limited by, the accompanying figures in which like numerals indicate similar elements.

(2) FIG. 1 illustrates an example communication network on which various features described herein may be used.

(3) FIG. 2 illustrates an example computing device that can be used to implement any of the methods, servers, entities, and computing devices described herein.

(4) FIGS. 3A-C illustrate an example concept diagram showing the removal of selected frames from a video stream.

(5) FIGS. 4A-B illustrate an example algorithm for removing frames from a video stream.

DETAILED DESCRIPTION

(6) In the following description of various illustrative embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments in which aspects of the disclosure may be practiced. It is to be understood that other embodiments may be utilized, and structural and functional modifications may be made, without departing from the scope of the present disclosure.

(7) FIG. 1 illustrates an example communication network **100** on which many of the various features described herein may be implemented. Network **100** may be any type of information distribution network, such as satellite, telephone, cellular, wireless, etc. One example may be an optical fiber network, a coaxial cable network, or a hybrid fiber/coax distribution network. Such networks **100** use a series of interconnected communication links **101** (e.g., coaxial cables, optical fibers, wireless, etc.) to connect multiple premises **102** (e.g., businesses, homes, consumer dwellings, etc.) to a local office or headend **103**. The local office **103** may transmit downstream information signals onto the links **101**, and each premises **102** may have a receiver used to receive and process those signals.

(8) There may be one link **101** originating from the local office **103**, and it may be split a number of times to distribute the signal to various premises **102** in the vicinity (which may be many miles) of the local office **103**. The links **101** may include components not illustrated, such as splitters, filters, amplifiers, etc. to help convey the signal clearly, but in general each split introduces a bit of signal degradation. Portions of the links **101** may also be implemented with fiber-optic cable, while other portions may be implemented with coaxial cable, other lines, or wireless communication paths. By running fiber optic cable along some portions, for example, signal degradation may be significantly minimized, allowing a single local office **103** to reach even farther with its network of links **101** than before.

(9) The local office **103** may include an interface, such as a termination system (TS) **104**. More specifically, the interface **104** may be a cable modem termination system (CMTS), which may be a

computing device configured to manage communications between devices on the network of links **101** and backend devices such as servers **105-107** (to be discussed further below). The interface **104** may be as specified in a standard, such as the Data Over Cable Service Interface Specification (DOCSIS) standard, published by Cable Television Laboratories, Inc. (a.k.a. CableLabs), or it may be a similar or modified device instead. The interface **104** may be configured to place data on one or more downstream frequencies to be received by modems at the various premises **102**, and to receive upstream communications from those modems on one or more upstream frequencies.

(10) The local office **103** may also include one or more network interfaces **108**, which can permit the local office **103** to communicate with various other external networks **109**. These networks **109** may include, for example, networks of Internet devices, telephone networks, cellular telephone networks, fiber optic networks, local wireless networks (e.g., WiMAX), satellite networks, and any other desired network, and the network interface **108** may include the corresponding circuitry needed to communicate on the external networks **109**, and to other devices on the network such as a cellular telephone network and its corresponding cell phones.

(11) As noted above, the local office **103** may include a variety of servers **105-107** that may be configured to perform various functions. For example, the local office **103** may include a push notification server **105**. The push notification server **105** may generate push notifications to deliver data and/or commands to the various premises **102** in the network (or more specifically, to the devices in the premises **102** that are configured to detect such notifications). The local office **103** may also include a content server **106**. The content server **106** may be one or more computing devices that are configured to provide content to users at their premises. This content may be, for example, video on demand movies, television programs, songs, text listings, etc. The content server **106** may include software to validate user identities and entitlements, to locate and retrieve requested content, to encrypt the content, and to initiate delivery (e.g., streaming) of the content to the requesting user(s) and/or device(s).

(12) The local office **103** may also include one or more application servers **107**. An application server **107** may be a computing device configured to offer any desired service, and may run various languages and operating systems (e.g., servlets and JSP pages running on Tomcat/MySQL, OSX, BSD, Ubuntu, Redhat, HTML5, JavaScript, AJAX and COMET). For example, an application server may be responsible for collecting television program listings information and generating a data download for electronic program guide listings. Another application server may be responsible for monitoring user viewing habits and collecting that information for use in selecting advertisements. Yet another application server may be responsible for formatting and inserting advertisements in a video stream being transmitted to the premises **102**. Although shown separately, one of ordinary skill in the art will appreciate that the push server **105**, content server **106**, and application server **107** may be combined. Further, here the push server **105**, content server **106**, and application server **107** are shown generally, and it will be understood that they may each contain memory storing computer executable instructions to cause a processor to perform steps described herein and/or memory for storing data.

(13) An example premises **102a**, such as a home, may include an interface **120**. The interface **120** can include any communication circuitry needed to allow a device to communicate on one or more links **101** with other devices in the network. For example, the interface **120** may include a modem **110**, which may include transmitters and receivers used to communicate on the links **101** and with the local office **103**. The modem **110** may be, for example, a coaxial cable modem (for coaxial cable lines **101**), a fiber interface node (for fiber optic lines **101**), twisted-pair telephone modem, cellular telephone transceiver, satellite transceiver, local wi-fi router or access point, or any other desired modem device. Also, although only one modem is shown in FIG. 1, a plurality of modems operating in parallel may be implemented within the interface **120**. Further, the interface **120** may include a gateway interface device **111**. The modem **110** may be connected to, or be a part of, the gateway interface device **111**. The gateway interface device **111** may be a computing device that

communicates with the modem(s) **110** to allow one or more other devices in the premises **102a**, to communicate with the local office **103** and other devices beyond the local office **103**. The gateway **111** may be a set-top box (STB), digital video recorder (DVR), computer server, or any other desired computing device. The gateway **111** may also include (not shown) local network interfaces to provide communication signals to requesting entities/devices in the premises **102a**, such as display devices **112** (e.g., televisions), additional STBs or DVRs **113**, personal computers **114**, laptop computers **115**, wireless devices **116** (e.g., wireless routers, wireless laptops, notebooks, tablets and netbooks, cordless phones (e.g., Digital Enhanced Cordless Telephone—DECT phones), mobile phones, mobile televisions, personal digital assistants (PDA), etc.), landline phones **117** (e.g. Voice over Internet Protocol—VoIP phones), and any other desired devices. Examples of the local network interfaces include Multimedia Over Coax Alliance (MoCA) interfaces, Ethernet interfaces, universal serial bus (USB) interfaces, wireless interfaces (e.g., IEEE 802.11, IEEE 802.15), analog twisted pair interfaces, Bluetooth interfaces, and others.

(14) FIG. 2 illustrates general hardware elements that can be used to implement any of the various computing devices discussed herein. The computing device **200** may include one or more processors **201**, which may execute instructions of a computer program to perform any of the features described herein. The instructions may be stored in any type of computer-readable medium or memory, to configure the operation of the processor **201**. For example, instructions may be stored in a read-only memory (ROM) **202**, random access memory (RAM) **203**, removable media **204**, such as a Universal Serial Bus (USB) drive, compact disk (CD) or digital versatile disk (DVD), floppy disk drive, or any other desired storage medium. Instructions may also be stored in an attached (or internal) hard drive **205**. The computing device **200** may include one or more output devices, such as a display **206** (e.g., an external television), and may include one or more output device controllers **207**, such as a video processor. There may also be one or more user input devices **208**, such as a remote control, keyboard, mouse, touch screen, microphone, etc. The computing device **200** may also include one or more network interfaces, such as a network input/output (I/O) circuit **209** (e.g., a network card) to communicate with an external network **210**. The network input/output circuit **209** may be a wired interface, wireless interface, or a combination of the two. In some embodiments, the network input/output circuit **209** may include a modem (e.g., a cable modem), and the external network **210** may include the communication links **101** discussed above, the external network **109**, an in-home network, a provider's wireless, coaxial, fiber, or hybrid fiber/coaxial distribution system (e.g., a DOCSIS network), or any other desired network. Additionally, the device may include a location-detecting device, such as a global positioning system (GPS) microprocessor **211**, which can be configured to receive and process global positioning signals and determine, with possible assistance from an external server and antenna, a geographic position of the device.

(15) The FIG. 2 example is a hardware configuration, although the illustrated components may be implemented as software as well. Modifications may be made to add, remove, combine, divide, etc. components of the computing device **200** as desired. Additionally, the components illustrated may be implemented using basic computing devices and components, and the same components (e.g., processor **201**, ROM storage **202**, display **206**, etc.) may be used to implement any of the other computing devices and components described herein. For example, the various components herein may be implemented using computing devices having components such as a processor executing computer-executable instructions stored on a computer-readable medium, as illustrated in FIG. 2. Some or all of the entities described herein may be software based, and may co-exist in a common physical platform (e.g., a requesting entity can be a separate software process and program from a dependent entity, both of which may be executed as software on a common computing device).

(16) One or more aspects of the disclosure may be embodied in a computer-usable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects,

components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other data processing device. The computer executable instructions may be stored on one or more computer readable media such as a hard disk, optical disk, removable storage media, solid state memory, RAM, etc. As will be appreciated by one of skill in the art, the functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects of the disclosure, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein.

(17) FIGS. 3A-C are conceptual visualizations of video streams containing sequences of image frames. In FIG. 3A, a video stream **301** is comprised of a plurality of image frames **302** that are sequentially displayed at a predetermined display rate (e.g., 30 frames per second). The encoding and delivery of the stream **301** may be done in a variety of ways. One example uses MPEG-2 (Moving Pictures Expert Group) encoding, which uses motion vector-based compression to efficiently represent the stream of image frames **302**. Using this compression, each frame is categorized as either an independent frame or a dependent frame. Independent frames are represented in the stream by data that is sufficient to generate the frame's complete image without knowledge about neighboring frames in the stream, similar to how a still image picture may be represented. The first frame after a scene change is typically represented using an independent frame.

(18) Dependent frames, as their name implies, are represented by data that is dependent on another frame in the stream, such as a corresponding independent frame, to generate the complete image of the dependent frame. The data representing a dependent frame may simply indicate changes with respect to a prior frame. For example, the data for a dependent frame may simply indicate that a first portion of the image remains unchanged from the prior frame, and that a second portion moves some distance (e.g., 3 pixels) to the right. In this manner, the data representing the dependent frame can omit the full details for the first portion of the dependent frame's image, thereby reducing the amount of data that is needed to be transmitted.

(19) In the MPEG-2 standard, which is one example audiovisual standard usable herein, independent frames are referred to as Intra-coded picture frames (I-frames), while dependent frames are referred to as either Predicted picture frames (P-frames), or a Bi-directional predicted picture frames (B-frames). A P-frame is dependent on a prior frame in the stream, while a B-frame is dependent on both a prior and a subsequent frame in the stream.

(20) In some embodiments, this motion vector-based compression can be done on an even smaller scale than a frame. For example, a single image frame may be divided into smaller sub-frames, and the same I/P/B-frame treatment may be performed at the sub-frame level. For example, the sub-frame can be an 8×8 collection of pixels, a 16×16 collection of pixels, or any other desired portion of the frames. In MPEG-2, so-called "macroblocks" may be formed by performing discrete cosine transforms on block portions of the frame. There, a 16×16 macroblock may be comprised of 16×16 luma (Y) samples and 8×8 chroma (Cb and Cr) samples.

(21) As noted above, a user's video device (e.g., gateway, smartphone, Digital Video Recorder, Set-Top Box, etc.) may need to tune to different channels (e.g., different Quadrature Amplitude Modulated (QAM) channels on different frequencies, different logical streams carried on a single QAM channel, different Internet Protocol streams, etc.) when presenting the user with video commercials during a break in scheduled programming (e.g., a television program, movie, etc.), and may need to return to an original channel or stream when the commercial break ends and the scheduled programming resumes. To allow smoother and quick transitions, the commercial break content may be shortened to allow moments of silent audio and blank and/or black video at the beginning and end of the commercials. So, for example, if the commercial stream **301** is 30-

seconds in duration, but the content provider wishes to have 0.5 seconds of silence/black on either end of the commercial, then the 30-second commercial needs to be reduced to 29 seconds in duration. To support this using the features described herein, certain frames **303** (FIG. 3B) in the stream may be selected for removal based on how disruptive their removal would be. The resulting shortened stream **304** (FIG. 3C) may then be transmitted for tuning and presentation to the user at the appropriate time.

(22) FIGS. 4A and B illustrate an example process by which frames, such as the selected frames **303**, may be chosen for removal to create the shortened stream **304**. The process may be performed by any one or more computing devices capable of video delivery, and in the FIG. 1 example network, the process may be performed at least in part by the content server **106** computing device.

(23) Beginning in step **401**, the computing device may receive a video stream **301** (or a file) and determine the frame rate (m) of the original video stream **301**. This may be, for example, identified by information in the video stream **301** file. In step **402**, the computing device may determine the amount of time (t) that is to be removed from the video stream **301**. For example, a 30-second segment, such as a commercial, may need to be reduced by one second, to result in a 29-second commercial.

(24) In step **403**, the computing device may determine the number of frames (Z) that will need to be removed from the video stream **301** to achieve the desired reduction in time (t). This calculation may be as follows: ($Z=t*m$).

(25) In step **404**, the computing device may begin to process each frame (e.g., in a looping fashion) in the stream **301** and generate corresponding comparison data that will be used in the eventual frame selection. In general, the frame selection algorithm may seek to identify the frames whose removal would cause the least amount of disruption to the resulting video stream, so that the shortened stream **304** will provide as close an experience as possible to the original stream **301**. This selection algorithm may need to compare data, the comparison data, representing visual elements of each of the various frames in the stream **301** to make this selection.

(26) In one example embodiment, the selection of frames may be based on comparisons of values representing portions of the video frame, such as the DC coefficient values for subsampled blocks of the frame. In one example embodiment, luminance (Y) components of blocks (e.g., 8×8 luma blocks) in an image may be processed through a Discrete Cosine Transform (DCT), resulting in an array of values, such as luminance DC coefficient values, for subsampled blocks of the frame, and these values may be compared in the eventual selection of frames. The values may be stored in the video file according to some video formats, although in other video formats the values may be obtained by performing the DCT on other data representing the frame. The details of this processing are discussed further below with regard to step **407**.

(27) The loop beginning in step **404** may sequentially step through each frame in the stream **301**. In step **405**, the computing device may select the next frame for processing. This selection may include maintaining a record of the sequence of frames in the video stream **301**, identifying the ones that have been processed to generate the comparison data for that frame, and selecting the earliest frame in the sequence that has not yet been processed to generate its comparison data.

(28) In step **406**, the computing device may determine whether the selected frame is an independent frame, or I-frame. If the selected frame is an independent frame, then in step **407**, the computing device may proceed to generate the comparison data for that frame by first identifying portions, such as each 8×8 luma block of DCT coefficients, in the frame.

(29) The comparison data, as will be discussed below, may be used to compare successive frames to identify an amount of change between the two frames. In some embodiments, the comparison data may be the DC coefficients that result when luminance components of a macroblock (e.g., a luma block) are processed using a Discrete Cosine Transform (DCT). Although the luminance component is used herein as an example, other features of an image frame may be used as well. For example, a chrominance component may be used instead.

(30) In step **408**, the computing device may perform a process such as a Discrete Cosine Transform (DCT) on the portion of the frame (e.g., an 8×8 pixel luma block). This step may be omitted, however, if the source video is in a compressed format that already includes DCT-transformed luma block information. In step **409**, the computing device may then store the DC coefficient of the result of the DCT for each portion (e.g., the luma DCT block). As a result, the computing device may store an array identifying the frame content, such as the luminance component DCT DC coefficients for all luma blocks in the frame. If the original frame is 720×480 pixels, and the luma blocks represent 8×8 pixel blocks, then this may result in a 90×60 array of luminance component DCT DC coefficients for the frame. Although DCT processes are used as an example above, other processes may be used to represent the content of a particular portion (or the entirety) of a frame for comparison purposes.

(31) In step **406**, if the selected frame is a dependent frame, then the luminance values for the various macroblocks in the frame may depend on neighboring frame(s), and may first need to be decoded before they can be processed to obtain the coefficients discussed above. Step **410** may begin a looping process to ensure that the luminance component coefficient values for all of the luma blocks in the current frame are decoded and available for use in the step **409** storing of the component (e.g., the DC component) from each luma block. The looping process may sequentially process each macroblock in the frame, and in step **411**, the computing device may select the next macroblock for processing.

(32) In step **412**, the computing device may determine whether the selected macroblock is an independent macroblock. As described above, frames may be independent or dependent, based on whether the frame's data can be derived without reference to a neighboring frame's data. The same approach may be used at the macroblock level. An independent macroblock may be represented by data that can be used to generate the macroblock's portion of the frame, while a dependent macroblock may be represented by data that refers to one or more macroblocks in reference pictures.

(33) If the macroblock is dependent on another macroblock, then in step **413**, the computing device may determine what other macroblock(s), or their predicted macroblock(s), are needed to decode the current macroblock. In step **414**, the computing device may retrieve the data for those other predicted macroblocks, and may use that data to decode the DC component information for each 8×8 luma block in the current macroblock. The process may then return to step **410**, and repeat until the computing device stores an array of data, such as the luminance component DCT DC coefficients, for the frame. At this point, the computing device now stores sufficient information for the dependent frame to generate its display without further reference to neighboring frame(s). From there, the process may proceed to step **407**, to generate the comparison data, e.g., the luminance DC component, for each of the macroblocks in the frame, and the two loops above may continue processing until the computing device has generated comparison data values, such as luminance component DCT DC coefficient values, for each frame in the video.

(34) Returning to step **404**, when all frames have been processed to generate the comparison data, the computing device may proceed to step **415**, and begin a process of comparing frames to select frames for removal. FIG. **4b** illustrates this portion of the process. In step **415**, the scene change value $C(v)$ for each frame may be calculated as a sum of the differences in the comparison values (e.g., the luminance DC component values) between corresponding sampling points in the frame and the immediately preceding frame. The determination for $C(v)$ of a v th image frame in a sequence of image frames may be expressed as follows:

(35)
$$C(v) = \sum_{n=1}^{n=h} \sum_{m=1}^{m=w} \text{abs}[DCv(m, n) - DCu(m, n)]$$
 wherein $C(v)$ =the scene change value for the v th frame in the video content, $DCv(m,n)$ =the Discrete Cosine Transform DC component of the m,n 8×8 block of DCT coefficients in the v th frame of the video content, $DCu(m,n)$ =the Discrete Cosine Transform DC component of the m,n 8×8 block of DCT coefficients in the u th

frame of the video content, $v=u+1$, w =the width of the frame, measured in 8×8 pixel blocks (8×8 pixel blocks are an example, and in alternate embodiments any desired sample size may be used), and h =the height of the frame, measured in 8×8 pixel blocks.

(36) In this example, the video content may be a sequence of image frames. As individual image frames are removed, the same calculations may be made for the image frames in the remaining sequence of image frames.

(37) The first frame may have its scene change value compared against zero values, resulting in a relatively high value of change.

(38) From step **415**, the computing device may determine and optionally store in memory a scene change value $C(v)$ for each frame in the video, and may proceed to step **416**, in which the various frames may be grouped into groups of pictures (GOP). In video coding, a GOP may be a collection of frames that have a common independent frame. For example, one GOP may comprise an independent frame and all of its corresponding dependent frames.

(39) In step **417**, the computing device may create a ranked frame list that ranks the various frames according to their scene change values $C(v)$, from lowest to highest.

(40) In step **418**, the computing device may then create a GOP round list that, at first, is simply a copy of the ranked frame list. The ranked frame list and GOP round list may be used in the ensuing steps to remove the frames that have the lowest scene change value, but to also evenly (to the extent possible) distribute the removal of frames across the various GOPs, so that no single GOP becomes disproportionately affected by having too many of its frames removed.

(41) In step **419**, the computing device may examine the GOP round list, and identify the first one on the list (i.e., the frame with the lowest scene change value $C(v)$) for removal. In step **420**, the selected frame may then be removed from the ranked frame list.

(42) In step **421**, the computing device may determine the GOP to which the selected frame belonged, and may remove all of the GOP's other frames from the GOP round list.

(43) The removal of the selected frame now means that the scene change value $C(v)$ calculated for the frame that followed the selected frame in the source video, the next frame, is outdated (since that scene change value calculated a difference using the now-removed frame's DC coefficient values). So in step **422**, the computing device may recompute the scene change value $C(v)$ for that next frame, but instead of comparing the next frame's coefficient values with those of the selected frame, the computing device may compare the values of the next frame with the frame that preceded the selected frame in the source video. In some embodiments, this recomputing may be optional, since the small change in scene often means that the recomputed value will be nearly the same as before. The recomputing may be skipped if the scene change value is smaller than a predetermined minimum change value, and this optional skipping may help reduce processing demand.

(44) In step **423**, the computing device may determine whether it has removed the desired number of frames (Z) from the source video. If it has not yet removed enough frames, then in step **424**, the computing device may determine whether the GOP round list is empty. If the GOP round list is not empty, then the process may return to step **419** to select the next frame for removal. If the GOP round list was empty in step **424**, then the computing device may proceed to step **425**, and copy the current ranked frame list (which omits the frames that have been removed so far in the process) to create a new GOP round list, and the process may then return to step **419** to select the next frame for removal.

(45) When the necessary number of frames (Z) has been removed, then the computing device may proceed to step **426**, and encode a new video file containing just the frames remaining in the ranked frame list, and may generate new time stamps for the resulting image frames. In step **427**, the computing device may encode a new audio soundtrack, based on the audio for the original video, to accompany the reduced set of frames in the video. The encoding of the new audio soundtrack may simply involve skipping portions of audio that accompanied the removed frames.

(46) In step **428**, the computing device may add black frames and silent audio portions to the beginning and end of the new video file. This may involve, for example, adding a number (Z) of frames equal to the number of removed frames, and the addition may be evenly split between the beginning and end of the new video file.

(47) In step **429**, the computing device may then take the new video file, and transmit it to receiving user devices instead of the original video file. This may entail, for example, remultiplexing the new video file in other streams or channels according to the same schedule used for the original video file.

(48) Although example embodiments are described above, the various features and steps may be combined, divided, omitted, rearranged, revised and/or augmented in any desired manner, depending on the specific outcome and/or application. Various alterations, modifications, and improvements will readily occur to those skilled in art. For example, the example process above uses luminance components, while other embodiments may use chrominance components instead. The example above also limits the frame removal to one frame per GOP per round. That limit can be revised to allow more than one frame from a GOP to be removed per round.

(49) Additional alterations, modifications, and improvements as are made obvious by this disclosure are intended to be part of this description though not expressly stated herein, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description is by way of example only, and not limiting. This patent is limited only as defined in the following claims and equivalents thereto.

Claims

1. A system comprising: a first computing device comprising: first one or more processors; and first memory storing first instructions; and a second computing device comprising: second one or more processors; and second memory storing second instructions; wherein the first instructions of the first memory, when executed by the first one or more processors, cause the first computing device to: shorten a segment by: removing, based on change information associated with the segment, an interior portion of the segment; and selecting, based on an image frame of the segment having a least amount of scene change compared to a preceding image frame, the image frame; generate, based on appending blank content to a beginning or an end of the shortened segment, an updated version of the segment; and send, to the second computing device, the updated version of the segment; and wherein the second instructions of the second memory, when executed by the second one or more processors, cause the second computing device to: receive the updated version of the segment.
2. The system of claim 1, wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to shorten the segment by: selecting, based on the change information, one or more image frames for removal from the segment.
3. The system of claim 1, wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to shorten the segment by: identifying a plurality of groups of pictures (GOPs) in the segment; and selecting, based on the change information and in one or more rounds, image frames for removal from the segment, wherein the selecting, in each round of the one or rounds, is limited to one image frame for each GOP of the plurality of GOPs.
4. The system of claim 1, wherein the blank content comprises one or more blank image frames.
5. The system of claim 1, wherein the change information comprises scene change values of a plurality of image frames in a sequence of image frames.
6. The system of claim 1, wherein the change information comprises scene change values of a plurality of image frames in a sequence of image frames, and wherein the first instructions of the

first memory, when executed by the first one or more processors, further cause the first computing device to: generate the scene change values according to a following formula:

$$C(v) = \text{Math.}_{n=1}^{n=h} \cdot \text{Math.}_{m=1}^{m=w} \text{abs}[DCv(m, n) - DCu(m, n)]$$
 wherein: $C(v)$ =a scene change value for a v th image frame, $DCv(m,n)$ =a Discrete Cosine Transform (DCT) DC component of an m,n 8×8 block of DCT coefficients in the v th image frame, $DCu(m,n)$ =a DCT DC component of an m,n 8×8 block of DCT coefficients in a u th image frame, $v=u+1$, w =a width of an image frame, and h =a height of an image frame.

7. The system of claim 1, wherein a duration of the interior portion is based on a duration of audio content to be appended.

8. The system of claim 1, wherein the blank content comprises audio content indicating silence.

9. The system of claim 1, wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to shorten the segment by: selecting, based on the change information, audio content for removal from the segment.

10. The system of claim 1, wherein the interior portion of the segment comprises one or more image frames, and wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to: skip audio content that accompanies the one or more image frames.

11. The system of claim 1, wherein the blank content comprises audio content indicating silence and corresponds to one or more blank image frames.

12. The system of claim 1, wherein the segment comprises an advertisement.

13. A system comprising: a first computing device comprising: first one or more processors; and first memory storing first instructions; and a second computing device comprising: second one or more processors; and second memory storing second instructions; wherein the first instructions of the first memory, when executed by the first one or more processors, cause the first computing device to: receive an audio content segment associated with a sequence of image frames; remove, based on change information associated with the audio content segment, a portion of the audio content segment occurring after a beginning of the audio content segment and prior to an end of the audio content segment, wherein the change information indicates changes in video images between sequential frames of the sequence of image frames; add, prior to the beginning of the audio content segment or after the end of the audio content segment, a blank portion comprising silent audio; and send the audio content segment to the second computing device; and wherein the second instructions of the second memory, when executed by the second one or more processors, cause the second computing device to: receive the audio content segment.

14. The system of claim 13, wherein the audio content segment is associated with the sequence of image frames, and wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to: add one or more blank image frames to the sequence of image frames.

15. The system of claim 13, wherein a duration of the audio content segment is based on a duration of the blank portion.

16. The system of claim 13, wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to select the removed portion by: determining a pair of sequential frames, in the sequence of image frames, having a least quantity of image change between the pair of sequential frames; and determining that the removed portion corresponds to the pair of sequential frames.

17. The system of claim 13, wherein the change information comprises scene change values of a plurality of image frames in the sequence of image frames, and wherein the first instructions of the first memory, when executed by the first one or more processors, cause the first computing device to: generate the scene change values according to a following formula:

$C(v) = \sum_{n=1}^{n=h} \sum_{m=1}^{m=w} \text{abs}[DCv(m, n) - DCu(m, n)]$ wherein $C(v)$ =a scene change value for a v th image frame, $DCv(m,n)$ =a Discrete Cosine Transform (DCT) DC component of an m,n 8×8 block of DCT coefficients in the v th image frame, $DCu(m,n)$ =a DCT DC component of an m,n 8×8 block of DCT coefficients in a u th image frame, $v=u+1$, w =a width of an image frame, and h =a height of an image frame.

18. A system comprising: a first computing device comprising: first one or more processors; and first memory storing first instructions; and a second computing device comprising: second one or more processors; and second memory storing second instructions; wherein the first instructions of the first memory, when executed by the first one or more processors, cause the first computing device to: shorten a sequence of image frames by: removing, from an interior portion of the sequence of image frames and based on scene change values of image frames of the sequence of image frames, one or more of the image frames; and selecting, from the sequence of image frames, an image frame having a least amount of scene change compared to a preceding image frame; generate an updated sequence of image frames by appending: one or more blank frames to a beginning or an end of the shortened sequence of image frames; and additional content proximate to the one or more blank frames; and send, the second computing device, the updated sequence of image frames; and wherein the second instructions of the second memory, when executed by the second one or more processors, cause the second computing device to receive the updated sequence of image frames.

19. The system of claim 18, wherein the sequence of image frames comprises an advertisement.

20. The system of claim 18, wherein the additional content comprises a video program.

21. The system of claim 18, wherein the sequence of image frames is associated with audio content, and the wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to: update the audio content by removing a portion, of the audio content, associated with the removed one or more of the image frames; and send the updated audio content with the updated sequence of image frames.

22. The system of claim 18, wherein the first instructions of the first memory, when executed by the first one or more processors, cause the first computing device to shorten the sequence of image frames by: identifying a plurality of groups of pictures (GOPs) in the sequence of image frames; and selecting, based on the scene change values and in one or more rounds, the one or more of the image frames, wherein the selecting, in each round of the one or rounds, is limited to one image frame for each GOP of the plurality of GOPs.

23. The system of claim 18, wherein the first instructions of the first memory, when executed by the first one or more processors, further cause the first computing device to: generate the scene change values according to a following formula:

$C(v) = \sum_{n=1}^{n=h} \sum_{m=1}^{m=w} \text{abs}[DCv(m, n) - DCu(m, n)]$ wherein: $C(v)$ =a scene change value for a v th image frame, $DCv(m,n)$ =a Discrete Cosine Transform (DCT) DC component of an m,n 8×8 block of DCT coefficients in the v th image frame, $DCu(m,n)$ =a DCT DC component of an m,n 8×8 block of DCT coefficients in a u th image frame, $v=u+1$, w =a width of an image frame, and h =a height of an image frame.

24. The system of claim 18, wherein the additional content comprises silent audio.

25. An apparatus comprising: one or more processors; and memory storing instructions that, when executed by the one or more processors, configure the apparatus to: shorten a segment by: removing, based on change information associated with the segment, an interior portion of the segment; and selecting, based on an image frame of the segment having a least amount of scene change compared to a preceding image frame, the image frame; generate, based on appending blank content to a beginning or an end of the shortened segment, an updated version of the segment; and send the updated version of the segment.

26. The apparatus of claim 25, wherein the instructions, when executed by the one or more processors, further cause the apparatus to: shorten the segment by: identifying a plurality of groups of pictures (GOPs) in the segment; and selecting, based on the change information and in one or more rounds, image frames for removal from the segment, wherein the selecting, in each round of the one or rounds, is limited to one image frame for each GOP of the plurality of GOPs.
27. The apparatus of claim 25, wherein a duration of the interior portion is based on a duration of audio content to be appended.
28. The apparatus of claim 25, wherein the blank content comprises audio content indicating silence.
29. The apparatus of claim 25, wherein the instructions, when executed by the one or more processors, further cause the apparatus to: shorten the segment by: selecting, based on the change information, audio content for removal from the segment.
30. A non-transitory computer-readable medium having instructions stored thereon that, when executed, cause: shortening a segment by: removing, based on change information associated with the segment, an interior portion of the segment; and selecting, based on an image frame of the segment having a least amount of scene change compared to a preceding image frame, the image frame; generating, based on appending blank content to a beginning or an end of the shortened segment, an updated version of the segment; and sending the updated version of the segment.
31. The non-transitory computer-readable medium of claim 30, wherein the instructions, when executed, further cause: shortening the segment by: identifying a plurality of groups of pictures (GOPs) in the segment; and selecting, based on the change information and in one or more rounds, image frames for removal from the segment, wherein the selecting, in each round of the one or rounds, is limited to one image frame for each GOP of the plurality of GOPs.
32. The non-transitory computer-readable medium of claim 30, wherein a duration of the interior portion is based on a duration of audio content to be appended.
33. The non-transitory computer-readable medium of claim 30, wherein the instructions, when executed, further cause: wherein the blank content comprises audio content indicating silence.
34. The non-transitory computer-readable medium of claim 30, wherein the instructions, when executed, further cause: shortening the segment by: selecting, based on the change information, audio content for removal from the segment.
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